# Yash Raj Shrivastava

Full-Stack Developer, Robotics & AI/ML Engineer | B.Tech CSE, IILM University (Class of 2029)
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## Profile

Computer-science undergraduate (IILM University, Gurugram) working across the full stack of product and systems engineering: full-stack web applications, embedded/robotics hardware (ESP32, ROS2), and applied AI/ML & GenAI. I take projects from idea to shipped artifact — 3D-printed and soldered hardware, open-source firmware, cloud-deployed software — and I document the work publicly. Currently co-building a stealth-mode SaaS, maintaining an open-source quadruped-robot codebase, and completing an 8-DOF robotics lab project (viva: 14.5/15). Daily practice includes local-LLM experimentation (Ollama/LM Studio), internal automation via agent workflows, and a second-brain knowledge-graph architecture that drives agent personalization.

## Education

- **IILM University, Gurugram** — B.Tech, Computer Science & Engineering (2025–2029)
- **Relevant lab coursework:** IoT & Robotics Foundation Lab (CSE25309) — ESP32 microcontrollers, I2C/PWM servo control, Tinkercad circuit simulation, 8-DOF quadruped project design and fabrication; Data Structures & OOPS laboratory sequences.

## Experience

**Co-founder & Product Designer — Stealth-Mode SaaS (Reservation & Analytics)** **(ongoing)**

- Leading product strategy, technical architecture, and investor/GTM outreach for a reservation and analytics SaaS platform.
- Own the full product lifecycle: ideation, UX design, full-stack development (Next.js, Prisma/PostgreSQL), and deployment.

**Lead Developer — Gurtai Bot (Local-LLM AI Assistant)** **(ongoing)**

- Built a Python-based Telegram AI assistant with local LLM inference via Ollama and LM Studio — privacy-first, zero per-query API cost.
- Designed a plugin-based modular architecture for extensible workflow automation and developer-centric tasks; containerized with Docker for consistent deployment.

**SIH 2026 (ISRO) — PS171 Browser Agent** **Optimization & Testing Lead**

- Worked on a PII-aware browser agent (WXT/React extension + backend) for ISRO's Smart India Hackathon; led the optimization and testing track: E2E test automation, performance tuning, and live-run diagnostics.
- Produced judge-facing project documentation and demo assets for the final submission.

**360 Labs — AI Test Work Week** **(Aug 2026)**

- Completed an intensive 1-week AI development program: a dozen rapid technical decisions, shipping fast under time pressure with validated output.

**Sagar Defence Engineering — Technical Interview (Drone/AI Internship)** **(Aug 2026)**

- Interviewed on robotics projects, embedded systems, and defence-domain requirements; assessed the defence hiring and clearance process.

## Software Projects

**CamPass — Student Safety & Tracking** **(Flutter, Node.js, PostgreSQL, FCM)**

- Location tracking and safety application: real-time geofence tracking, SOS alerts, and smart notifications via Firebase Cloud Messaging.
- Digital pass system with barcode scanning and role-based access control.

**Machina (Stealth SaaS) — Web Platform** **(Next.js, Prisma, Vercel)**

- Reservation and analytics platform: schema design with Prisma/PostgreSQL, component-driven UI, deployed to Vercel; ongoing iteration on GTM strategy (target: CA-firm and SME owners).

## AI & ML Projects

**RiceScan — AI Crop-Disease & Quality Detection** **(Python, EfficientNetV2S, OpenCV)**

- Full vision pipeline: custom camera with controlled lighting → image capture → segmentation → classification → quality-percentage estimation.
- Integrated a local knowledge base for symptoms and treatment plans; designed a user-feedback loop to retrain the model.
- Impact: quality control for farmers and food organizations.

**Gurtai — Local-LLM Agent Stack** **(Ollama, LM Studio, Docker)**

- Local inference via Ollama/LM Studio: cost control, privacy, and offline capability; model-selection and quantization tradeoffs; plugin system for workflow automation; Docker deployment.
- Prompt-engineering and LLM-ops practice from daily use; DeepLearning.AI certified (Anthropic rich-context apps).

## Hardware & Robotics Projects

**Sesame Robot — ESP32 8-DOF Quadruped (Open Source)** **github.com/Yashop965/sesame-robot**

- Built a mini quadruped from scratch: ESP32 DevKitC-32E, 8× MG90S metal-gear servos, 3D-printed PLA chassis (Fusion 360).
- Custom firmware: Wi-Fi AP with captive-portal browser control, REST API (`/api/status`, `/api/command`), OLED face expressions (SSD1306).
- Pinned ESP32Servo v3.0.9; allocated PWM timers 0–3 across all 8 servos; mapped the full GPIO pinout across all legs.
- **Incident work:** 8×500 mA servo draws produced 4 A peaks that destroyed the original buck converter; root-caused via measurement and $I^2R$ thermal modeling; shipped an ECO firmware actuation limit (max 4 concurrent servos, 200 ms stagger) and upgraded to a 5 A buck with heatsink — verified safe at 2 A.
- Status: assembly complete, basic walk gait achieved, Gazebo (ROS2) simulation debugging in WSL2. Total project cost ≈ INR 3,150.

**Swarm Drone System — Carrier Hexacopter Architecture** **(ROS2 Humble, Gazebo, MQTT)**

- Designed a carrier swarm: a hexacopter mother drone carries 4–6 autonomous sub-drones, delegates tasks via capability-aware scoring, and manages autonomous docking after missions.
- Full ROS2 Humble workspace: URDF hexacopter model, motor-controller node, task router with battery/proximity/sensor scoring, mesh-network protocol (ROS→MQTT/TCP/UDP bridge).
- Headless Gazebo physics simulation: 4/4 task assignments validated, clean colcon build in 1.29 s.
- Full stack: ground-station UI → mother drone → sub-drone wireless mesh → return-and-dock cycle with failsafe autonomous navigation.

**Other Hardware**

- **RiceScan hardware:** custom camera rig with controlled lighting for grain-image acquisition.
- **DIY electronics & optics:** exposure-control systems and custom camera prototypes (sensors, lenses, microcontrollers).
- **Drone components:** ESC (30 A, Hobbywing, DShot) and motor-driver testing for flight stability, power efficiency, and compatibility.

## Experiments, Automation & Ops

- **Local AI inference:** Ollama and LM Studio — running, quantizing, and evaluating local LLMs for agent experiments and Gurtai; prompt/context design and model-selection tradeoffs.
- **Internal automation:** personalised Hermes Agent workflows for internal automation — documentation, research digests, CI triage, and outreach pipelines; a personal second brain (Obsidian) with a knowledge-graph architecture that drives agent personalization and optimization.
- **Architectural thinking:** data-model design (Prisma/PostgreSQL schemas), plugin/modular system design (Gurtai), multi-agent and fleet topologies, and simulation-first hardware development (WSL2 + Gazebo).
- **Simulation:** ROS2 Humble + Gazebo headless testing in WSL2 — URDF modeling, gait-pattern validation (trot/walk/pronk), task-assignment verification.
- **Ops:** Docker for reproducible environments (Gurtai and dev fleets); CI/CD failure triage (local-pass/CI-fail diagnosis); E2E test automation with Playwright; headless-server deployments (WSL2).
- **Experimentation:** power-distribution experiments on legged robots (buck-converter thermal limits, contact-resistance losses), PWM timer allocation studies, I2C bus topology (OLED 0x3C, PCA9685 0x40).

## Skills

- **Languages:** Python, JavaScript, TypeScript, C++, Dart, SQL, Bash
- **AI & ML:** GenAI applications, prompt engineering, LLMOps, AI agents & chatbots, RAG concepts, OpenCV, EfficientNet (transfer learning), local inference (Ollama, LM Studio)
- **Web & Mobile:** Next.js, React, WXT (browser extensions), Node.js, Express, Flutter, Prisma/PostgreSQL, Firebase, Docker
- **Embedded & Hardware:** ESP32 (firmware, PWM, I2C), Raspberry Pi, Arduino, ROS2 Humble, Gazebo, servo control, soldering and schematic-level circuit work, 3D printing (Fusion 360)
- **Ops & Tooling:** Git, GitHub, Docker, CI/CD debugging, Playwright E2E, Vercel, ESLint, WSL2, AWS (basics: EC2/S3/Bedrock concepts), custom agent workflows (Hermes), Obsidian second brain / knowledge graph
- **Lab & EE fundamentals:** Tinkercad circuit simulation, breadboard design, sensors/actuators (PIR, IR, HC-SR04, 7-segment, 16×2 LCD-I2C), AC/DC and Ohm's law

## Certifications

- DeepLearning.AI — *AI Python for Beginners*; *Build Rich-Context AI Apps with Anthropic*
- 360 Labs — AI Test Work Week (Aug 2026)

## Open Source & Links

- Open-source quadruped firmware and hardware: github.com/Yashop965/sesame-robot
- GitHub: github.com/Yashop965 | Resume (web): yashop965.github.io/resume